

IPv6 Implementation -- Guided Notes ANSWER KEY

Unit 2.3.3 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

For instructor use / distribute after students complete the notes.

Question 1

IPv6 uses _____ -bit addresses, compared to IPv4's 32-bit addresses. Written in hexadecimal, an IPv6 address has _____ groups of four hex digits separated by colons. The total number of possible IPv6 addresses is approximately 3.4×10^{38} (340 undecillion).

Answer: 128-bit; 8 groups; approximately 3.4×10^{38} (340 undecillion).

Question 2

Two rules for abbreviating IPv6 addresses: (1) Leading _____ in any group may be omitted. (2) One consecutive run of all-zero groups may be replaced with _____. This compression can only be used _____ per address.

Answer: Zeros (leading zeros); :: (double colon); once.

Question 3

IPv6 eliminates the need for NAT because every device can receive a globally unique _____ address. This restores the internet's original design principle of _____ connectivity.

Answer: Global unicast; end-to-end.

Question 4

Match each IPv6 address type to its description:

Global Unicast -> _____

Link-Local -> _____

Multicast -> _____

Loopback -> _____

Answer: Global Unicast: publicly routable, starts with 2000::/3. Link-Local: valid only on the local segment, starts with FE80::/10, auto-configured. Multicast: one-to-many delivery, starts with FF00::/8. Loopback: ::1/128, equivalent to 127.0.0.1.

Question 5

SLAAC stands for _____. It allows IPv6 hosts to configure their own _____ automatically using the network prefix advertised by a router, combined with the host's _____ or a random value -- without needing a DHCP server.

Answer: Stateless Address Autoconfiguration; global unicast address; EUI-64 (MAC address).

Question 6

IPv6 replaces ARP with _____. It uses _____ messages to discover the link-layer address of a neighbor. This is part of the _____ Discovery Protocol (NDP).

Answer: Neighbor Discovery; Neighbor Solicitation and Neighbor Advertisement; Neighbor.

Question 7

During the transition from IPv4 to IPv6, what is a dual-stack configuration? What is tunneling in the context of IPv6 transition, and name one common tunneling mechanism.

Answer: Dual-stack: a device runs both IPv4 and IPv6 simultaneously, using whichever is supported by the destination. Tunneling: IPv6 packets are encapsulated inside IPv4 packets to traverse IPv4-only networks. Common mechanisms include 6to4, Teredo, and ISATAP.

Question 8 -- Real World Application

REAL WORLD: A technician pings a server and receives a reply from FE80::1. What does this tell the technician about the current state of connectivity? What would need to be true for the server to be reachable from the public internet over IPv6?

Answer: FE80:: is a link-local address -- the reply confirms Layer 3 IPv6 connectivity on the local segment only. Link-local addresses are not routable beyond the local link. For the server to be reachable from the public internet, it needs a global unicast address (2000::/3 range), and the upstream router must advertise the correct prefix and have IPv6 routing configured.